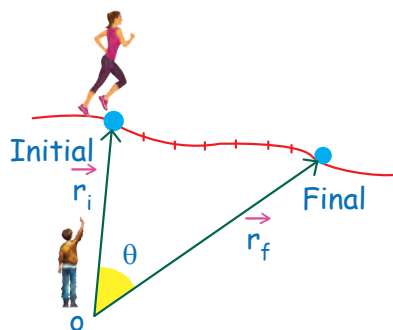


पूरी mechanics में circular motion सबसे ज्यादा important chapter है जिसका use WPE, COM, Rotation, Electrostatic, Magnetism etc. में होगा..... बहुत ही आराम से सुकून से basic fundamental and definition अच्छे से पढ़ना क्योंकि ये आपका base ready करेगा..... are you ready



Let us feel a beautiful girl is running in front of you and you are watching her.... अबे feel कर ना.... बोले तो Let a particle is moving in a random curved path as shown in fig. and we are observing it from O.



SKC

Bhai teri mundi kitna angle ghumi is called angular displacement kis rate se ghumi is called angular velocity of radius vector.



★ Angular position is the angle (θ) made by the position vector of a point (measured from a fixed origin) with respect to a chosen reference line or direction.

★ $\theta \rightarrow$ angular displacement $\Rightarrow$ angle rotated by radius vector or (position vector).

★ Difference between the two angular positions of the particle moving along any arbitrary path w.r.t. some fixed point is called angular displacement.

or

★ Angular velocity (ω) = $\frac{d}{dt}$ = Rate of change of angular position (for radius vector).

★ Angular acc. ' α ' = Rate of change of angular velocity $\frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$

★ Angular variables: θ , ω , α depend on observer.

Average Angular Speed

It is the ratio of total angular displacement and the time taken to do so.

$$\omega_{av} = \frac{\text{Total angular displacement}}{\text{Total time taken}} = \frac{\Delta\theta}{\Delta t}$$

Instantaneous Angular Velocity

It is the rate of change of angular position at some particular instant.

Instantaneous angular velocity

$$\bar{\omega} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\bar{\theta}}{\Delta t} = \bar{\omega} = \frac{d\bar{\theta}}{dt}$$

★ Instantaneous angular velocity can also be called simply as angular velocity.

Q. If angular displacement is given as

$$\theta = 2t^2 + 3 \text{ where 't' is time then}$$

(i) Find out average angular speed upto 3 sec.

(ii) Angular velocity, angular acceleration at 3 sec.

Sol. (i) $\omega_{\text{avg}} = \frac{\text{Total angular displacement}}{\text{total time}} = \frac{\theta_f - \theta_i}{t_2 - t_1}$

$\theta_f = 2(3)^2 + 3 = 21 \text{ rad} \quad (\text{at } t = 3 \text{ sec})$

$\theta_i = 2(0) + 3 = 3 \text{ rad} \quad (\text{at } t = 0 \text{ sec})$

So, $\omega_{\text{avg}} = \frac{21-3}{3} = 6 \text{ rad/sec}$

(ii) $\omega_{\text{instantaneous}} = \frac{d\theta}{dt} = 4t$

$\omega_{\text{at } t=3 \text{ sec}} = 4 \times 3 = 12 \text{ rad/sec}$

$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} = 4 \text{ rad/sec}^2$

भाई θ , ω , α के बीच बिल्कुल वही relation है जो kinematics में x , v , a के बीच हुआ करता था.....

$$\begin{aligned} x &\rightarrow \theta \\ v = \frac{dx}{dt} &\rightarrow \omega = \frac{d\theta}{dt} \\ a = \frac{dv}{dt} &\rightarrow \alpha = \frac{d\omega}{dt} \end{aligned}$$

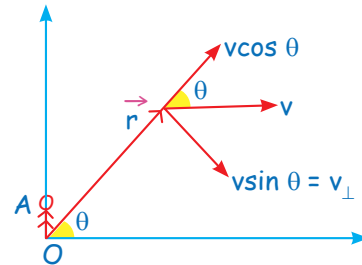
सर मैं तो circular motion पढ़ने आया था वहाँ पर θ , α , ω होता था लेकिन आपने तो यहाँ पर किसी random path के लिए चाहे वो straight line हो उसके लिए θ , α , ω define कर दिए



बेटा बिल्कुल कर सकते हैं ये समझ ले कि circular motion इन सबका special case ही तो है



★ Suppose a particle is moving parallel to x-axis on xy plane with constant velocity $\vec{v}$ if observer is at origin as shown in fig. below. Then angular velocity of radius vector w.r.t origin is given by

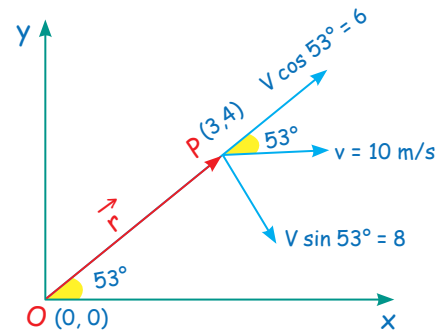


$\frac{d\theta}{dt} = \omega = \text{ang. vel. of Position vector (P.V)}$

$\omega = \frac{v \sin \theta}{r} = \frac{v_{\perp}}{r}$
Magnitude

Q. A particle is moving with constant velocity 10 m/s along positive x-axis. Find ω of radius vector about origin when particle is at (3, 4).

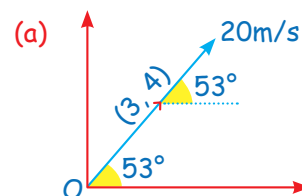
Sol.



$(\omega_p)_{\text{abt 'O'}} = \frac{V \sin \theta}{r} = \frac{8}{5} \text{ rad/s}$

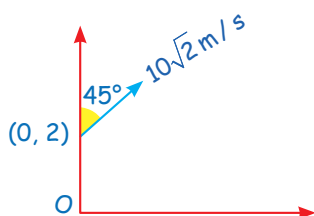
$\omega = \frac{\vec{r} \text{ के } \perp \text{ वाली vel.}}{r}$

Q. Find ω of radius vector about 'O' in following question.



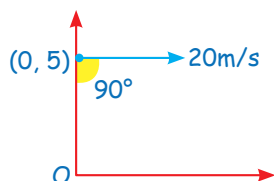
Sol. $\omega = 0$

(b)



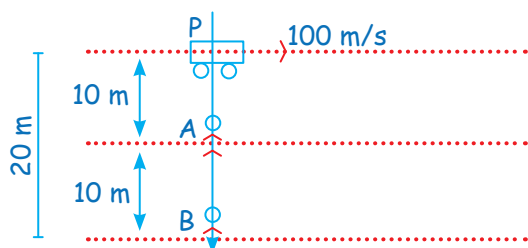
Sol. $\omega = \frac{10}{2} = 5 \text{ rad/s}$

(c)



Sol. $\omega = \frac{20}{5} = 4 \text{ rad/s}$

(d)

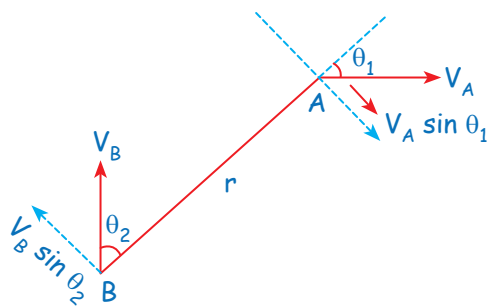


Sol. $(\omega_P)_{\text{wrt 'A'}} = \frac{100}{10} = 10 \text{ rad/s}$

$(\omega_P)_{\text{wrt 'B'}} = \frac{100}{20} = 5 \text{ rad/s}$

Calculation of $\omega_{A/B}$ When A and B Both are Moving

Relative angular velocity of a particle 'A' with respect to the other particle 'B' is the angular velocity of the position vector of 'A' with respect to 'B'. Which means that it is the rate at which position vector of 'A' with respect to 'B' rotates at that instant.



$$\omega_{AB} = \frac{(V_{AB})_{\perp}}{r_{AB}}$$

Here $(V_{AB})_{\perp}$ = Relative velocity $\perp$ to position vector AB

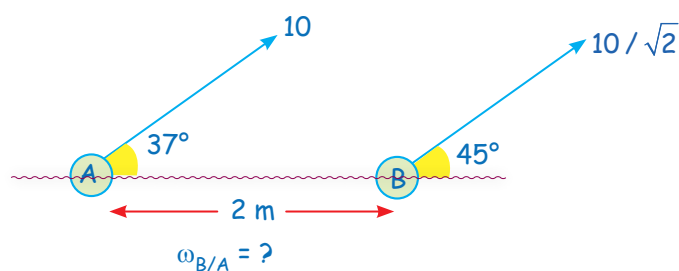
r_{AB} = separation between A and B.

$$(V_{AB})_{\perp} = V_A \sin \theta_1 + V_B \sin \theta_2$$

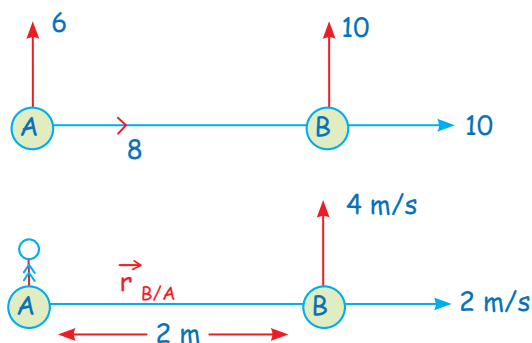
$$r_{AB} = r$$

$$\omega_{AB} = \frac{V_A \sin \theta_1 + V_B \sin \theta_2}{r}$$

Q. Calculate angular velocity of B wrt A.



Sol. Breaking in components



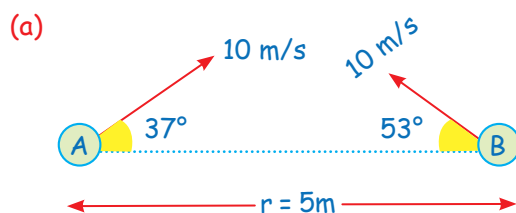
$$\vec{\omega}_{B/A} = \frac{4}{2} = 2 (\text{ACW}) = 2\hat{k}$$

SKC

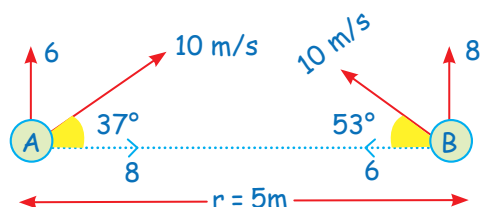
* for $\omega_{B/A}$

- Sabse pehle dono particle ko ek line se jor do.
- Dono vel. ko, line ki taraf aur line k $\perp^{\text{ar}}$ tod lo.
- Kisi ek particle ke uper jake beth jao [relative]
- Ghumane wali vel. / r [divide] kro.

Q. Find $\omega_{B/A}$

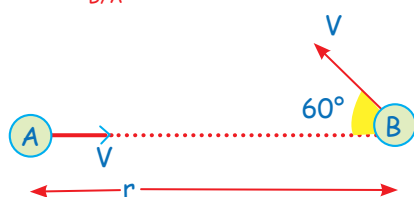


Sol.

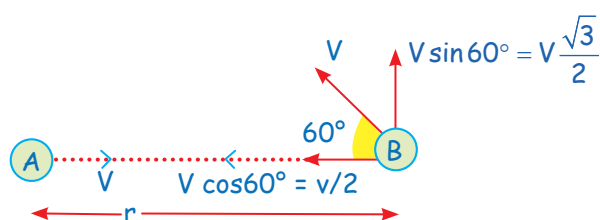


$$\omega_{B/A} = 2/5 \text{ (ACW)}$$

(b) Find $\omega_{B/A}$



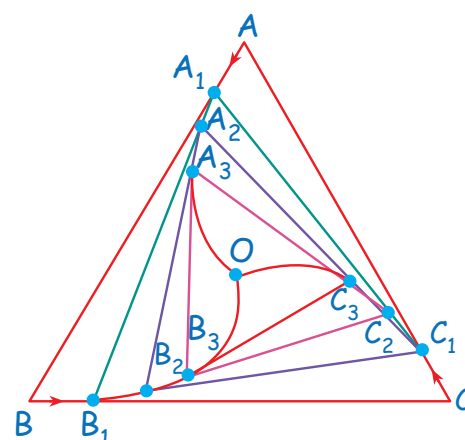
Sol.



$$\omega_{B/A} = \frac{V\sqrt{3}/2}{r}$$

Q. Three particles A, B and C are situated at the vertices of an equilateral triangle ABC of side ℓ at $t = 0$. Each of the particles moves with constant speed v . A always has its velocity along AB, B along BC and C along CA. At what time will the particles meet each other?

Sol. Roughly motion of the particles will be like as shown in fig. and by symmetry they will meet at the centroid.



Velocity of A is v along AB. The velocity of B is along BC. Hence rate of decrease of separation

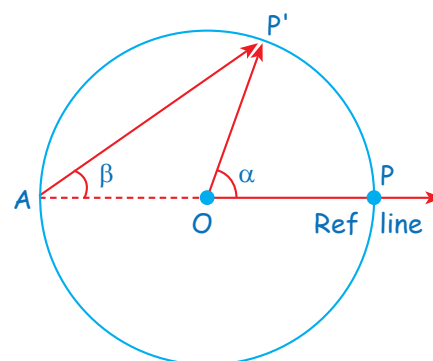
$$= v + v \cos 60^\circ = \frac{3v}{2} = \text{constant}$$

Time taken to reduce the separation from ℓ to zero.

$$t = \frac{\ell}{\frac{3v}{2}} = \frac{2\ell}{3v}$$

Relative Angular Velocity

Angular velocity is defined with respect to the point from which the position vector of the moving particle is drawn. Here angular velocity of the particle w.r.t. 'O' and 'A' will be different.



$$\omega_{PO} = \frac{d\alpha}{dt} = \omega \text{ (say);}$$

$$\omega_{PA} = \frac{d\beta}{dt} = \frac{\omega}{2} \quad \left[\because \beta = \frac{\alpha}{2} \right]$$

NORMAL ACCELERATION AND TANGENTIAL ACCELERATION

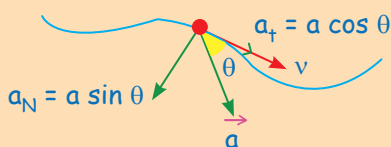
Suppose a particle is moving in a random curved path with variable speed then

Component of acceleration towards the velocity is called tangential acceleration and component acceleration perpendicular to the velocity is called normal acceleration.



काम का डब्बा

Tangential Acceleration



- ★ Component of $\vec{a}$ along vel. = $a \cos \theta$
= a_t = Tangential Acceleration

★ $a_t = a \cos \theta = \frac{\vec{a} \cdot \vec{v}}{v}$ (Magnitude)

- ★ In vector form

$$\vec{a}_t = a \cos \theta \hat{v} = \frac{\vec{a} \cdot \vec{v}}{v} \hat{v}$$

- ★ It is responsible to change the magnitude of $\vec{v}$

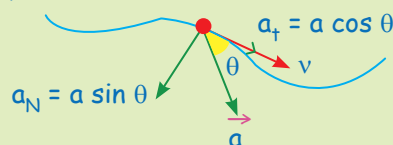
$$\frac{d(\text{speed})}{dt} = a_t$$

Rate of change of speed is tangential acceleration.

- ★ If $\vec{a}_t$ is in the direction $\times^n$ of $\vec{v} \Rightarrow$ speed up
- ★ If $\vec{a}_t$ is in the opp. direction of $\vec{v} \Rightarrow$ speed down.
- ★ If speed is constant $\Rightarrow a_t = 0$

Normal Acceleration

- ★ Component of acceleration perpendicular to the velocity.
- ★ It is responsible to change the direction of velocity.



- ★ Normal acceleration = $a_N = a \sin \theta$
- ★ $\vec{a}_N = \vec{a} - \vec{a}_t$ (very important)

SKC

- ★ Agar particle kisi curve par chal rha ho to, uski vel. Ki dirⁿ nikalne ke liye us curve per tangent Khicho.
- ★ Agar speed const. hai to tangential acc. 0 hoga, chaye Kesa bhi motion ho [1D, 2D, 3D]
- ★ Agar speed badli to tangential acc hai. Agar dirⁿ badli matlb normal acc. hai

Q. If a particle is moving such that it's acceleration velocity is given by

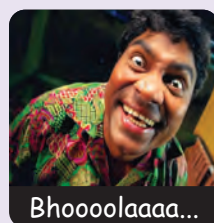
$$\vec{a} = 4\hat{i} - 3\hat{j}$$

$$\vec{v} = \hat{i} + \hat{j} \text{ Find } a_t \text{ and } a_N$$

Sol. $\vec{a}_t = \frac{\vec{a} \cdot \vec{v}}{v} \hat{v} = \left(\frac{(4\hat{i} - 3\hat{j}) \cdot (\hat{i} + \hat{j})}{\sqrt{2}} \right) \cdot \frac{(\hat{i} + \hat{j})}{\sqrt{2}} = \frac{\hat{i} + \hat{j}}{2}$

$$\vec{a}_N = \vec{a} - \vec{a}_t = (4\hat{i} - 3\hat{j}) - \frac{\hat{i} + \hat{j}}{2} = \frac{7\hat{i} - 7\hat{j}}{2}$$

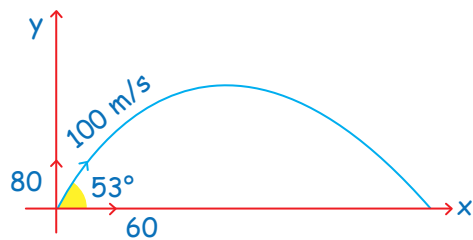
भाई आगे बढ़ने से पहले तुझे यह promise करना पड़ेगा की अगर तुम्हे velocity, acc vector form में दिए गए तो formula में value put करके a_t और a_N निकाल लोगे। अगर यहाँ अच्छे से practice नहीं की तो पता है ना exam में क्या होगा।



Bhoooooaaaaa...



Q. A particle is projected with velocity 100 m/s at an angle 53° with horizontal as shown in fig. Find $\vec{a}_t$ & $\vec{a}_N$ at $t = 2$ sec.



Sol. $t = 2, \vec{v} = 60\hat{i} + 60\hat{j}$

$$\vec{a} = -10\hat{j}$$

$$\vec{a}_t = \frac{\vec{a} \cdot \vec{v}}{v} \hat{v} = \frac{-600}{60\sqrt{2}} \cdot \frac{60\hat{i} + 60\hat{j}}{60\sqrt{2}}$$

$$\vec{a}_N = \vec{a} - \vec{a}_t$$

Now you can solve.

Q. If $\vec{V} = 3t^2\hat{i} + 4t\hat{j}$ at $t = 1$ sec find $\vec{a}_t$ & $\vec{a}_N$.

Sol. $\vec{v} = 3t^2\hat{i} + 4t\hat{j}, t = 1, \vec{v} = 3\hat{i} + 4\hat{j}$

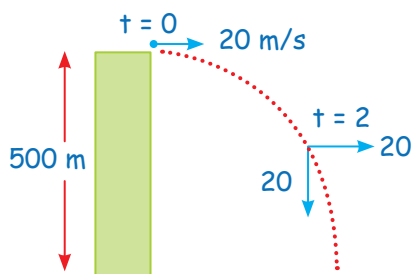
$$\vec{a} = 6t\hat{i} + 4\hat{j}, t = 1, \vec{a} = 6\hat{i} + 4\hat{j}$$

$$\vec{a}_t = \frac{(18+16)\hat{v}}{5} = \frac{34}{5} \left(\frac{3\hat{i} + 4\hat{j}}{5} \right)$$

$$\vec{a}_N = \vec{a} - \vec{a}_t = 6\hat{i} + 4\hat{j} - 34/5 \cdot \left(\frac{3\hat{i} + 4\hat{j}}{5} \right)$$

Now you can solve

Q. Find tangential acc. at $t = 2$ sec.

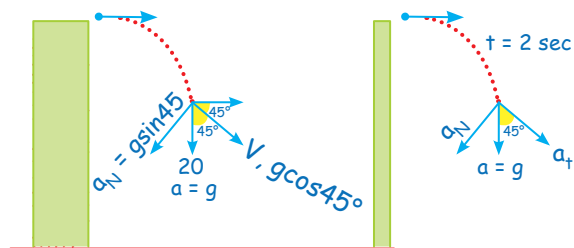


Sol. $\vec{v} = 20\hat{i} - 20\hat{j}$

$$\vec{a} = -10\hat{j}$$

$$\vec{a}_t = \frac{\vec{a} \cdot \vec{v}}{v} \hat{v} = \frac{0 + 200}{20\sqrt{2}} \times \left(\frac{20\hat{i} - 20\hat{j}}{20\sqrt{2}} \right) = 5\hat{i} - 5\hat{j}$$

OR



$$a_t = g \cos 45^\circ = \frac{10}{\sqrt{2}} = m/s^2$$

तो चलो अब शुरू करते हैं Circular motion....

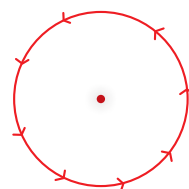
यही है ना... जिस के लिए पढ़ रहे थे। हम पहले ही बोले दे रहे हैं कि ये सबसे important chapter है जो WPE, COM, Rotation, Fluid, Electrostatic, Magnetism etc. सभी जगह use होगा।



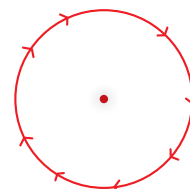
CIRCULAR MOTION

★ When a particle moves in a plane such that its distance from a fixed (or moving) point remains constant then its motion is called as circular motion with respect to that fixed point. That fixed point is called centre and the distance is called radius of circular path.

★ Direction of $\vec{\omega}$ is given by right hand thumb rule.



ACW (Sense)



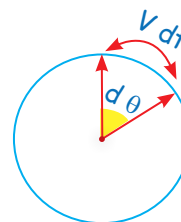
CW (Sense)

direction of $\vec{\omega} = \odot \hat{k}$,
 $\perp^r$ outward

direction of $\vec{\omega} = \otimes \hat{k}$,
 $\perp^r$ inwards

★ $\theta \rightarrow$ angular displacement For a complete revolution angular disp. is 2π but disp. = 0

$$\star \omega = \frac{d\theta}{dt}, \alpha = \frac{d\omega}{dt}$$



Uniform Circular Motion

- ★ Motion of a particle along the circumference of a circle with a constant speed is called uniform circular motion.
- ★ Angular Speed, (ω), constant
- ★ $a_t = 0$, $\alpha = 0$
- ★ It is an accelerated motion because direction of velocity vector keeps on changing.
- ★ The position vector and velocity vector keep on changing, though their magnitudes remain constant.
- ★ The acceleration of the particle points always towards center at each point on its path. Which is normal acc. or centripetal acc. Whose value is v^2/r , or $r\omega^2$
- ★ $\vec{r} \cdot \vec{v} = 0$ ★ $\vec{v} \cdot \vec{a} = 0$
- ★ $\omega = \frac{v}{R}$ (magnitude)
- ★ $v = R\omega$ $\vec{v} = \vec{\omega} \times \vec{r}$
- ★ For 1 complete revolution
Speed = Dist./time = $2\pi R/T = R\omega$
 $\omega = 2\pi/T$

Non-Uniform Circular Motion

Non-uniform circular motion

If the speed of the particle moving in a circle is not constant, the acceleration has both the radial and the tangential components.

Examples of non uniform circular motion

A merry-go-round spinning up from rest to full speed, or a ball whirling around in a vertical circle.

In non-uniform circular motion

Speed $|\vec{v}| \neq \text{constant}$

angular velocity $\omega \neq \text{constant}$

If at any instant

v = magnitude of velocity of particle

ω = angular velocity of particle,

then, at that instant $v = r\omega$

where r = radius of circular path

भाई θ, ω, α के बीच बिल्कुल वही relation है जो kinematics में x, v, a के बीच हुआ करता था.....

$$x \rightarrow \theta$$

$$v = \frac{dx}{dt} \rightarrow \omega = \frac{d\theta}{dt}$$

$$a = \frac{dv}{dt} \rightarrow \alpha = \frac{d\omega}{dt}$$

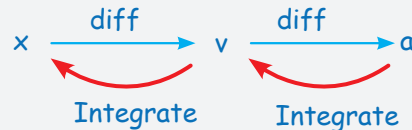
भाई रुक पहले इन्हे detail में अच्छे से digest कर



★ $\vec{v}, \vec{a} \Rightarrow$ Same sign (speed up)

★ $\vec{\omega}, \vec{\alpha} \Rightarrow$ Same sign (speed up)

Linear Kinematics	Circular Motion Kinematics
$v = \frac{dx}{dt}$	$\omega = \frac{d\theta}{dt}$
$\vec{a} = \frac{d\vec{v}}{dt}$	$\vec{\alpha} = \frac{d\vec{\omega}}{dt}$
$(x-t) \Rightarrow \text{slope} \Rightarrow \text{vel.}$	$(\theta-t) \Rightarrow \text{slope} \Rightarrow \omega$
$(v-t) \Rightarrow \text{slope} \Rightarrow \text{acc.}$	$(\omega-t) \Rightarrow \text{slope} \Rightarrow \alpha$
$(a-t) \Rightarrow \text{area} \Rightarrow \text{change in velocity}$	$(\alpha-t) \Rightarrow \text{area} \Rightarrow \omega$
$(v-t) \Rightarrow \text{area} \Rightarrow \text{disp.}$	$(\omega-t) \Rightarrow \text{area} \Rightarrow d\theta$



If $a \rightarrow \text{const.}$

$$v = u + at$$

$$S = ut + \frac{1}{2} at^2$$

$$v^2 = u^2 + 2as$$

$$s \rightarrow \theta$$

$$a \rightarrow \alpha$$

$$v \rightarrow \omega$$

If $\alpha \rightarrow \text{const.}$

$$\omega = \omega_i + \alpha t$$

$$\theta = \omega_i t + \frac{1}{2} \alpha t^2$$

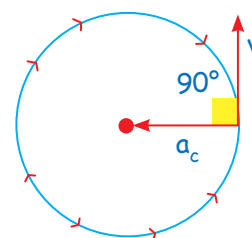
$$\omega^2 = \omega_i^2 + 2\alpha\theta$$

CENTRIPETAL ACCELERATION (NORMAL ACC.)

Component of acc. $\perp$ to the vel. If at any instant speed of particle performing circular motion is v then

Centripetal acc. (a_c)

$$= \frac{v^2}{R} \text{ (toward centre)}$$



Q. A particle is moving in a circular path of radius 4 m with speed 10 m/s in clockwise sense in x-y plane. Find ω , a_t , a_c

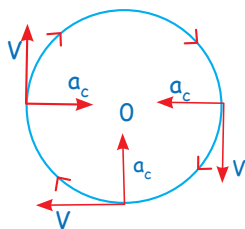
Sol. $\omega = \frac{V}{R}$

$\omega = 10/4$ or $\vec{\omega} = -2.5 \hat{k}$,

$a_t = 0$

$a_t = R\alpha \Rightarrow \alpha = 0$

$a_c = \frac{V^2}{R} = \frac{100}{4} = 25$ (towards centre)



Next ques. is very important ques.

अच्छे से एक-एक चीज निकालना सीख लेना।



Q. A particle is moving in a circular path of radius 2 m such that its speed changes as $v = 2t^2$. Find ω , α , a_t , a_c at $t = 2$ s and angle made by a_{net} with velocity at $t = 2$ s.

Sol. $a_t = \frac{d}{dt}(\text{speed}) = 4t$

at $t = 2 \Rightarrow v = 2 \times 4 = 8$ m/s

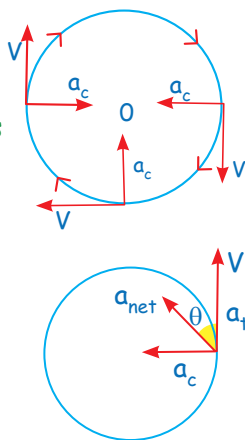
$\omega = \frac{V}{R} = \frac{8}{2} = 4$ rad/s

$a_c = \frac{v^2}{R} = \frac{8^2}{2} = 32$ m/s²

a_{net} = resultant of a_t & a_c

$= \sqrt{a_t^2 + a_c^2} = \sqrt{8^2 + 32^2}$

$\tan \theta = \frac{a_c}{a_t} = \frac{32}{8} = 4$



SKC

Agar particle circular motion Kar raha hai to a_c hoga hi hoga. Aur uski speed badal rehi hae to a_t hoga hi hoga. If speed up $\Rightarrow a_t \Rightarrow \vec{v}$ ki side If speed down $\Rightarrow a_t \Rightarrow \vec{v}$ ke opposite side. Jab bhi Kabhi circular motion me acceleration puchega matlab net acc. puch rha h.

Q. A particle is rotating in a circular path of radius 2 m such that angle rotated by (radius vector) is given as $\theta = t^3/6$.

Find ω , α , a_t , a_c , a_{net} , v at $t = 2$ sec.

Sol. $\theta = \frac{t^3}{6}$

$\omega = \frac{d\theta}{dt} = \frac{3t^2}{6} = \frac{t^2}{2}$

$\alpha = \frac{2t}{2} = t$

$t = 2, \omega = 2, \alpha = 2$

$a_t = R\alpha$

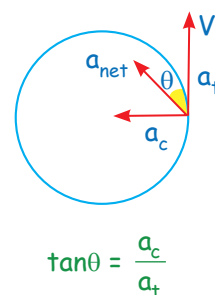
$a_t = 4$

$a_c = R\omega^2 = 2 \times 4 = 8$ m/s²

$a_{net} = \sqrt{4^2 + 8^2}$

$v = R\omega$

$v = R \frac{t^2}{2} = 2 \times \frac{4}{2} = 4$



Q. A particle travels in a circle of radius 20 cm at a speed that uniformly increases. If the speed changes from 5 m/s to 6 m/s in 2 s, find the angular acceleration.

Sol. The tangential acceleration is given by

$a_t = \frac{dv}{dt} = \frac{v_2 - v_1}{t_2 - t_1}$

$= \frac{6.0 - 5.0}{2} \text{ m/s}^2 = 0.5 \text{ m/s}^2$

The angular acceleration is $\alpha = a_t / r$

$= \frac{0.5 \text{ m/s}^2}{20 \text{ cm}} = 2.5 \text{ rad/s}^2$

Q. A particle is moving in a circular path with velocity varying with time as $v = 1.5t^2 + 2t$. If the radius of circular path is 2 cm, the angular acceleration at $t = 2$ sec will be

Sol. Given $v = 1.5t^2 + 2t$

Tangential acceleration $a_t = \frac{dv}{dt} = 3t + 2$

Now angular acceleration at time t :

$\alpha = \frac{a_t}{r} \Rightarrow \alpha = \frac{3t + 2}{2 \times 10^{-2}}$

Angular acceleration at $t = 2$ sec

$(\alpha)_{at \ t=2 \text{ sec}} = \frac{3 \times 2 + 2}{2 \times 10^{-2}} = \frac{8}{2} \times 10^2 = 4 \times 10^2$
 $= 400 \text{ rad/sec}^2$

- Q. The shaft of an electric motor starts from rest and on the application of a torque, it gains an angular acceleration given by $\alpha = 3t - t^2$ during the first 2 seconds after it starts after which $\alpha = 0$. The angular velocity after 6 sec will be

Sol. Given $\alpha = 3t - t^2$

$$\Rightarrow \frac{d\omega}{dt} = 3t - t^2 \Rightarrow d\omega = (3t - t^2)dt$$

$$\Rightarrow \omega = \frac{3t^2}{2} - \frac{t^3}{3} + c$$

$$\text{at } t = 0, \omega = 0$$

$$\therefore c = 0 \Rightarrow \omega = \frac{3t^2}{2} - \frac{t^3}{3}$$

Angular velocity at $t = 2$ sec,

$$(\omega)_{t=2 \text{ sec}} = \frac{3}{2}(4) - \frac{8}{3} = \frac{10}{3} \text{ rad/sec}$$

Since there is no angular acceleration after 2 sec

$\therefore$ The angular velocity after 2 sec remains the same.

- Q. Two particles A and B move anticlockwise with the same speed v in a circle of radius R and are diametrically opposite to each other. At $t = 0$, A is imparted a tangential acceleration of constant magnitude $a_t = \frac{72v^2}{25\pi R}$. Calculate the time in which A collides with B, the angle traced by A during this time, its angular velocity and radial acceleration of A at the time of collision.

Sol. $\omega_{0(\text{rel})} = 0, \theta_{\text{rel}} = \pi, \alpha_{\text{rel}} = \frac{72v^2}{25\pi R^2}$

$$\theta_{\text{rel}} = \omega_{0(\text{rel})} t + \frac{1}{2} \alpha_{\text{rel}} t^2$$

$$\pi = 0 + \frac{1}{2} \frac{72v^2}{25\pi R^2} t^2 \Rightarrow t = \frac{5\pi R}{6v}$$

$$\text{Angle traced by A, } \theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\theta = \frac{v}{R} \cdot \frac{5\pi R}{6v} + \frac{1}{2} \frac{72v^2}{25\pi R^2} \cdot \left(\frac{5\pi R}{6v}\right)^2$$

$$= \frac{5\pi}{6} + \pi = \frac{11}{6} \pi$$

$$\text{Angular velocity } \omega = \omega_0 + \alpha t = \frac{v}{R} + \frac{72v^2}{25\pi R^2} \cdot \left(\frac{5\pi R}{6v}\right)$$

$$= \frac{v}{R} + \frac{12v}{5R} = \frac{17v}{5R}$$

$$a_c = \omega^2 R = \left(\frac{17v}{5R}\right)^2 R = \frac{289v^2}{25R}$$

UNIFORM AND NON-UNIFORM CIRCULAR MOTION

Uniform Circular Motion	Non-uniform Circular Motion
Speed $\rightarrow$ const.	Speed $\rightarrow$ change up (speed or down)
$\omega \rightarrow$ const.	$\omega \rightarrow$ change
$a_{\text{net}} = \sqrt{a_t^2 + a_c^2}$	$a_t \neq 0, \alpha \neq 0$
$a_{\text{net}} = a_c = \frac{V^2}{R} = R\omega^2$	$a_{\text{net}} = \sqrt{a_t^2 + a_c^2}$

DYNAMICS OF CIRCULAR MOTION

SKC

अब जो article हम पढ़ने जा रहे हैं बहुत ही interesting है अब तक हमने देखा कोई particle अगर circular motion कर रहा है तो उसके पास a_c होता है तो भाई ये a_c ऐसे थोड़े ही अपने आप पैदा हो जाएगा..... अब कोई तो force particle पर लग रहे होंगे जिन्होंने a_c centripetal acc. दिया है some of all these forces towards the centre is called centripetal force.

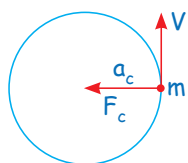
Centripetal Force

$$a_c = \frac{V^2}{R} = R\omega^2 \text{ (toward the centre)}$$

$$\text{Centripetal force} = ma_c = \frac{mv^2}{R} = mR\omega^2$$

Centripetal force लगने वाला force नहीं है So, block पर centripetal force centre की तरफ लगता है जैसे गधो वाले sentence use मत करना..... just say centripetal force is the vector sum of all the forces towards the centre acting on the particle performing circular motion.



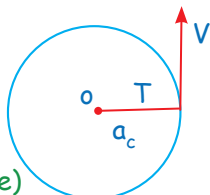


Q. A particle is moving on a smooth horizontal floor in a circular path with const. speed v , as shown in diagram find Tension

Sol. $mg = \perp^r$ inside

$N = \perp^r$ outside

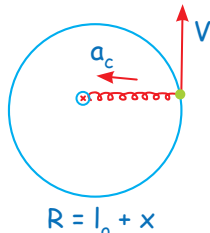
$$T = ma_c = \frac{mv^2}{R} \text{ (Towards centre)}$$



Q. A particle is moving on a smooth horizontal floor in a circular path with const. speed v , as shown in diagram. If natural length of spring is l_0 find elongation in spring.

$$\text{Sol. } Kx = \frac{mv^2}{R} = \frac{mv^2}{l_0 + x}$$

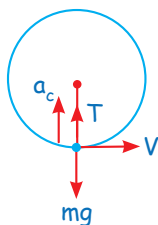
x : elongation



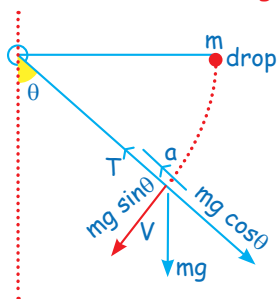
Q. A particle is performing circular motion in vertical plane such that its vel. at lowest point is v . Find tension at lowest point.

$$\text{Sol. } T - mg = m \times a_c = \frac{mv^2}{r}$$

$a_r = 0$



Q. A particle is released when string is horizontal as shown in fig. Find tension when string makes angle θ with verticle and having speed V .



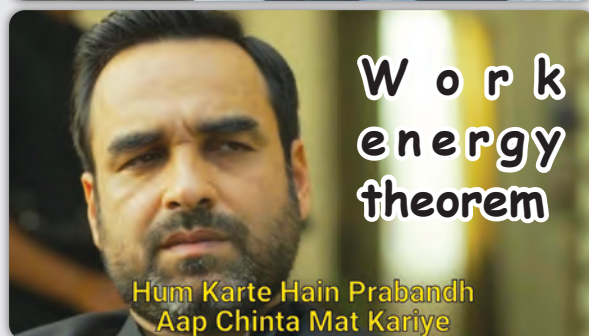
$$\text{Sol. } T - mg \cos \theta = \frac{mv^2}{R}$$

$$\text{or } T = mg \cos \theta + \frac{mv^2}{R}$$

$$mg \sin \theta = ma_t$$

$$a_t = g \sin \theta$$

Circular Motion



SKC



भाई circular motion 12th तक तेरे बहुत काम आएगा इसलिए नीचे 4 Saleemians Khopcha Concept लिख रहा हूँ इनसे तुम्हारी सारी problem solve हो जाएगी।

1. सबसे पहले यह देखो की centre कहाँ है।
2. Partical की FBD बनाओ।
3. सारे forces को centre की तरफ और उसके perpendicular तोड़लो।
4. Centre की तरफ वाले force का sum mv^2/r या $mr\omega^2$ के बराबर करदो

बस इन्हे अच्छे से apply करना सीख लेना इसके लिए बहुत सारे ques. add कर रहा हूँ।



Conical Pendulum

Q. Particle is performing circular motion in horizontal circle with constant angular speed ω as shown in fig.

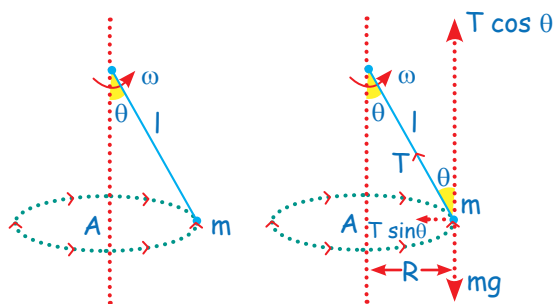
$\theta \rightarrow \text{Const.}, \text{ length of string} \rightarrow l$

$$\text{Sol. } T \cos \theta = mg$$

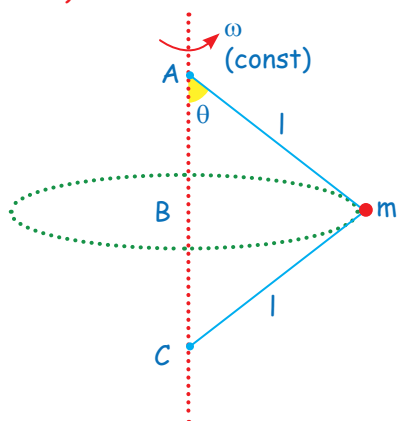
$$T \sin \theta = \frac{mv^2}{R} = mR\omega^2$$

$$l \sin \theta = R$$

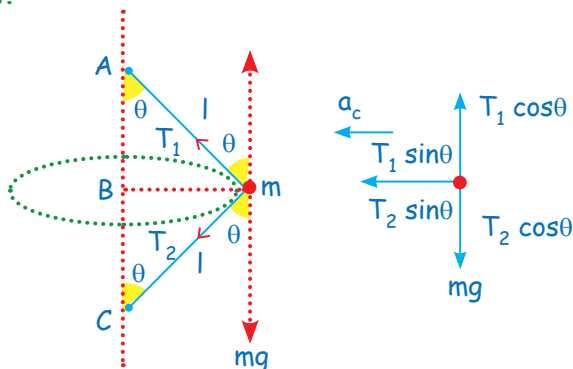
यह बहुत बड़ा बेशर्म ques है JEE MAINS मे तो यह बहुत बार आ चुका है।



Q. In following fig. particle m is moving in circular motion. Find tension in both the string (massless).



Sol.



$$T_1 \cos \theta = mg + T_2 \cos \theta$$

$$T_1 \sin \theta + T_2 \sin \theta = m r \omega^2$$

$$r = l \sin \theta$$

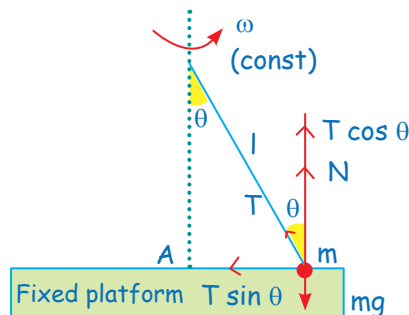
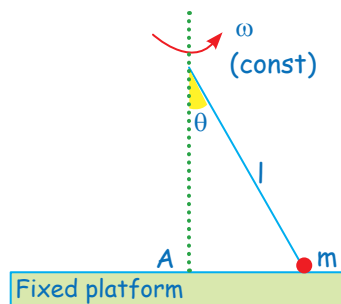
Now you can solve above eqn.

आप ये सिखो की circular motion की eqn कैसे लिख रहे है ques वाला कोई भी physical quantity पूछ सकता है It's just maths

Q. In following fig. a circular platform is rotating with constant ω on smooth horizontal surface, on the platform a ball is attached to the axis of platform by a thin rod of length l .

(a) Find normal force exerted by ball on the platform. ($\theta = 60^\circ$, $\omega = 10 \text{ rad/s}$, $l = 10 \text{ cm}$)

Sol.



$$T \cos \theta + N = mg$$

$$T \sin \theta = m r \omega^2$$

$$r = l \sin \theta$$

Solve and get $N = 5 \text{ newton}$

(b) In above ques. find $\omega_{\max}$ at which particle does not loose contact with platform. Find T in string at that condition.

Sol.

$$T \cos \theta + N = mg$$

$$T \cos \theta = mg \quad [N = 0]$$

$$T \sin \theta = m r \omega^2$$

Solve and get

$$\omega = \sqrt{g \tan \theta / r}$$

अबे कुछ नहीं करना बस circular motion की eqn $N = 0$ put करदो

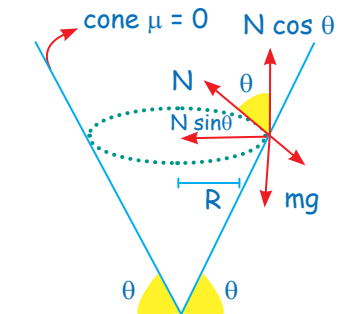


Q. Particle is moving with const. speed on the smooth frictionless surface (cone) as shown in fig in a circular path of radius R . find Normal force on particle?

Sol. $N \cos \theta = mg$

$$N \sin \theta = \frac{m v^2}{R}$$

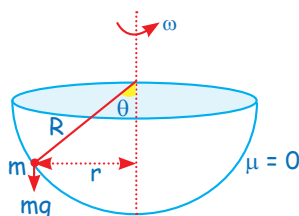
Now you can solve



देख रहा है Binod.... कैसे एक conical pendulum के concept का use 5-6 सवालो मे किया जा रहा है



- Q. Hemisphere bowl is rotating with const ω as shown in figure such that particle is at rest wrt bowl. (चिपकर धूम रहा है)
Find normal (N) force on particle.



Sol. Replica of conical Pendulum.

$$N \cos \theta = mg$$

$$N \sin \theta = mr\omega^2$$

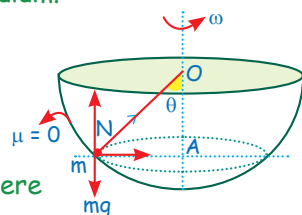
$$R \sin \theta = r$$

$R \rightarrow$ Radius of hemisphere

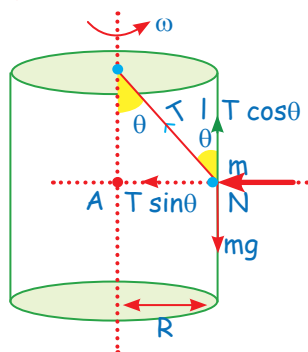
$r \rightarrow$ Radius of circle

Solve and get

$$N = mR\omega^2$$



- Q. Cylinder is rotating with const ω . Particle is at rest wrt cylinder. Write the releavent circular motion eqn



Sol. $T \cos \theta = mg$

$$T \sin \theta + N = mR\omega^2$$

$$= m(l \sin \theta) \omega^2$$

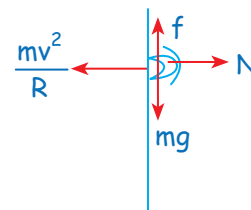
(l : length of string)

- Q. A motor cyclist is driving in a horizontal circle on the inner surface of vertical cylinder of radius R . Find out the minimum velocity for which the motorcyclist can do this. (Death Well)

Sol. $N = \frac{mv^2}{R}$

$$f = mg$$

$$f_{\max} = \frac{\mu mv^2}{R}$$

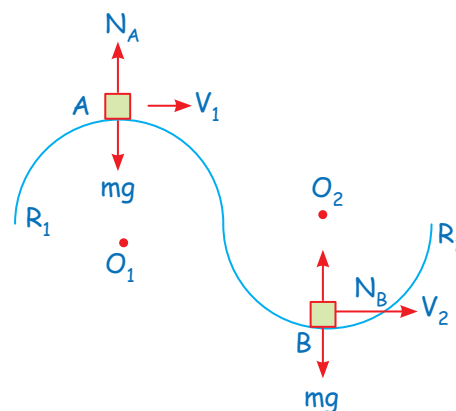


Cyclist does not drop down when

$$f_{\max} \geq mg \Rightarrow \frac{\mu mv^2}{R} \geq mg$$

$$v \geq \sqrt{\frac{gR}{\mu}}$$

- Q. Particle moving on a curved path as shown in vertical plane. Assume particle does not lose contact. Write the circular motion eqn when particle is at point A and B.

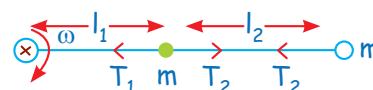


Sol.

$$mg - N_A = \frac{mV_1^2}{R_1} \quad ; \quad N_B - mg = \frac{mV_2^2}{R_2}$$

$$N_B > N_A$$

- Q. System is on the frictionless Horizontal floor, both masses are moving in a circular path with same ω . Find Tension in String.

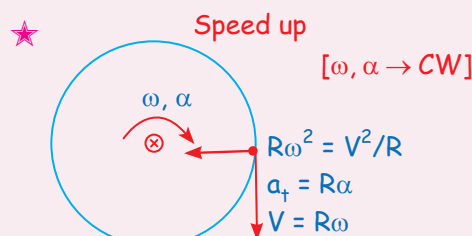
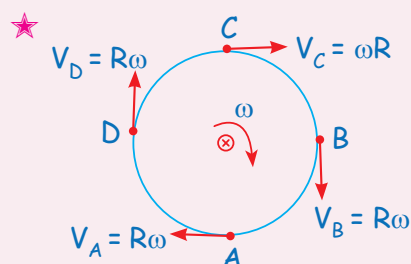
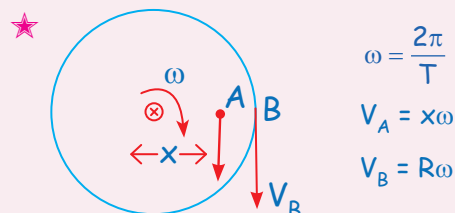




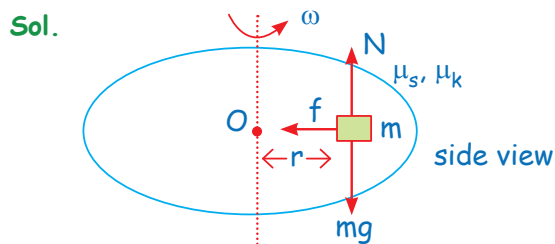
$$T_1 - T_2 = m l_1 \omega^2 \quad (\text{for first mass})$$

$$T_2 = m(l_1 + l_2) \omega^2 \quad (\text{for second mass})$$

Suppose a disk is rotating on a horizontal floor with ω then velocity of different points in following diagram will be



Q. Table of radius r_0 is rotating about an axis $\perp$ to the plane of table and passing through centre of table. If angular vel of table is ω const, A mass 'm' is placed at a distance 'r' from centre of table such that mass remains at rest wrt table, find friction force acting on block.

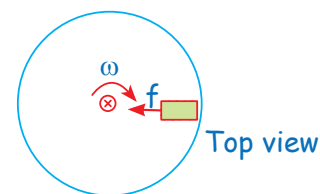


$$a_t = 0$$

$mg \rightarrow$ Inside

$N \rightarrow$ outside

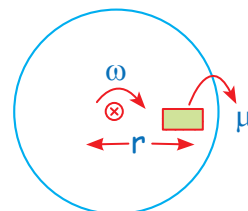
$$f_s = mr\omega^2$$



Q. Find $\omega_{\max}$ so that block does not slip

OR

Find $\omega_{\max}$ so there block remains at rest wrt disc



Sol. $mg \rightarrow$ Inside

$N \rightarrow$ Outside

$$a_t = 0$$

$$f_s = mr\omega^2$$



$$(f_s)_{\max} = \mu_s mg$$

$$\omega_{\max} = \sqrt{\frac{\mu_s g}{r}}$$

Q. Suppose in above ques, disk start rotating from rest having constant angular acc α find when block will slip if coefficient of friction is μ .

Sol. $f_{\text{net}} = ma_{\text{net}}$

$$f_{\text{net}} = m\sqrt{a_c^2 + a_t^2}$$

$$a_t = r\alpha$$

$$a_c = r\omega^2$$

When block is just about to slip. At this time

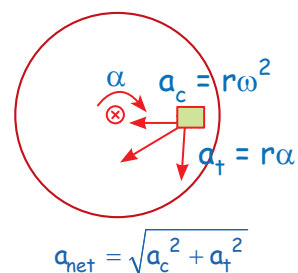
$$\omega = 0 + \alpha t = \alpha t$$

$$f = f_{\max} = \mu_s mg$$

$$\mu_s mg = m\sqrt{(r\omega^2)^2 + (r\alpha)^2}$$

$$\mu_s mg = m\sqrt{(r\alpha^2 t^2)^2 + (r\alpha)^2}$$

Now you can find t by putting value of μ_s, α, r, t





अबे value put करके इतना तो solve कर लोगे ना यहाँ ये important है की friction का एक component a_r दे रहा है और दुसरा component required centripetal force.

Q. Find the max speed that a vehicle (bicycle) can possess moving in a circle with const speed v such that surface is horizontal & coeff. of friction is μ .

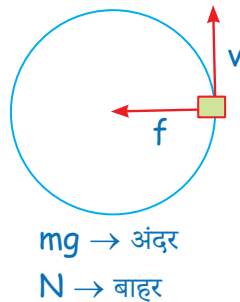
Sol.

$$f = \frac{mv^2}{R} \quad \text{max}$$

$$\mu N = \frac{mv_{\max}^2}{R}$$

$$\mu mg = \frac{mv_{\max}^2}{R}$$

$$V_{\max} = \sqrt{\mu g R}$$



Banking of Road



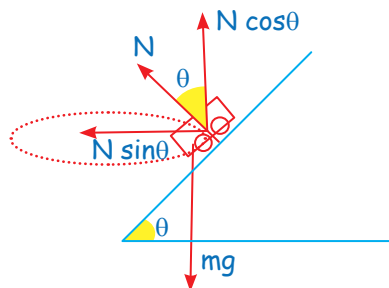
(i) Frictionless

$$N \cos \theta = mg$$

$$N \sin \theta = \frac{mv^2}{R}$$

$$\tan \theta = \frac{v^2}{Rg}$$

$$V = \sqrt{Rg \tan \theta}$$

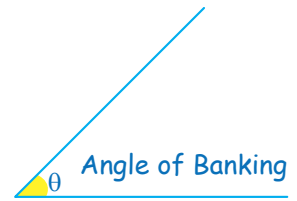


(ii) If friction is present

$$V_{\max} = \sqrt{Rg \tan(\theta + \phi)}$$

$$V_{\min} = \sqrt{Rg \tan(\theta - \phi)}$$

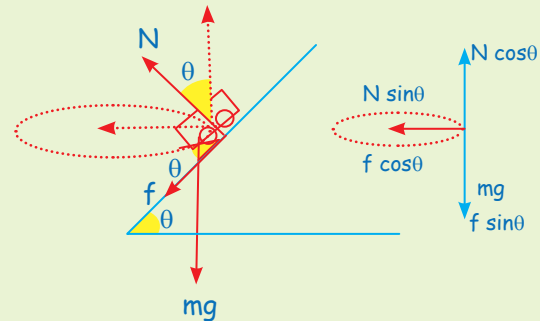
$$\text{where } \mu_s = \tan \phi$$



(ϕ : Angle of friction)

Calculation for $V_{\max}$:

$$N \cos \theta = mg + f \sin \theta$$



$$N \sin \theta + f \cos \theta = \frac{mv^2}{R}$$

$$\text{for } V_{\max} = (f_s)_{\max} = \mu_s N$$

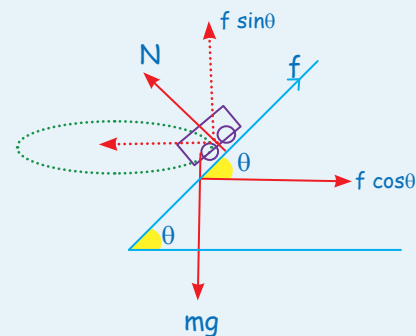
Solve and get

$$V_{\max} = \sqrt{Rg \tan(\theta + \phi)}$$

Calculation for $V_{\min}$:

Just Replace f to $-f$ in $V_{\max}$.

$$V_{\min} = \sqrt{Rg \tan(\theta - \phi)}$$



Q. Suppose in above article we have $R = 20\text{m}$, $\mu_s = 1/\sqrt{3}$, $\theta = 45^\circ$.

(a) Find max/min possible speed for which car turn safely

Sol. $V_{\max} = \sqrt{Rg \tan(\theta + \phi)} = \sqrt{20 \times 10 \tan(45 + 30)}$

$V_{\min} = \sqrt{Rg \tan(\theta - \phi)} = \sqrt{200 \tan(45^\circ - 30^\circ)}$

(b) At what velocity friction is not required to turns safely.

Sol. $V = \sqrt{Rg \tan \theta} = \sqrt{200 \tan 45^\circ} = \sqrt{200}$



काम का डब्बा

देख भाई JEE MAINS के लिए ये 4 formula बहुत काम के है exam से पहले इन्हे जरूर पढ़के जाना।

★ $V_{\max} = \sqrt{\mu g R}$ (Max speed on horizontal plane for safely turn of a bicycle)

★ $V = \sqrt{Rg \tan \theta}$ (banking of road when $\mu = 0$)

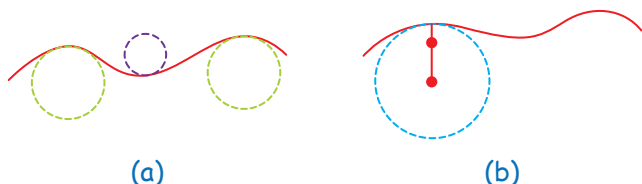
★ Banking of road when friction is present

$V_{\max} = \sqrt{Rg \tan(\theta + \phi)}$

$V_{\min} = \sqrt{Rg \tan(\theta - \phi)} \quad \tan \theta = \mu_s = \tan \phi$

Radius of Curvature (ROC)

Any curved path can be assume to be made of infinite circular arcs. ROC at a point is defined as the radius of that circular arc which fits at that particular point on the curve.



★ $a_n = \frac{v^2}{R} \Rightarrow R = \frac{v^2}{a_n} = \text{ROC}$

★ $\text{ROC} = R = \frac{v^2}{a_\perp}$

भाई कुछ नहीं करना बस सबसे पहले normal acc निकाल लो और v^2/a_n करदो यही ROC है और अगर $y = f(x)$ given है means particle moves in a trajectory given by $y = f(x)$ then ROC at any point (x, y) of the trajectory is given by

$$R = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$$



Q. A stone is projected with speed u and angle of projection is θ .

(a) Find radius of curvature at $t = 0$.

(b) Find ROC at highest point.

Sol. (a) At $t = 0$

$a_\perp = g \cos \theta,$

$R = \frac{v^2}{a_\perp} = \frac{u^2}{g \cos \theta}$

(b) At highest point $v_p = u \cos \theta$ and $a_\perp$ or $a_n = g$

Hence ROC at 'P' = $\frac{u^2 \cos^2 \theta}{g}$

